

Executive Summary – Gapminder Analysis

1. Project Overview

The Gapminder Analysis project uses Power BI to explore global patterns in population, life expectancy, and GDP per capita across countries and continents. The dashboard provides interactive visualizations that allow users to compare countries, analyze continental differences, and observe how these indicators change over time.

The report includes country-level and continent-level analysis, interactive year and continent slicers, KPI cards, a map, treemap, donut chart, bar chart, detailed table, and trend analysis. The detailed pages also provide year-wise trends for population, life expectancy, and GDP per capita.

2. Key Areas of Analysis

- **Population Analysis**

The dashboard analyzes population at both country and continent levels. The map, donut chart, treemap, and KPI visuals provide different views of population distribution.

The country-level visualizations help identify countries with larger population contributions, while the continental analysis provides a broader understanding of population distribution.

- **Life Expectancy Analysis**

The dashboard compares average life expectancy across continents and provides a detailed trend visualization.

The analysis allows users to observe how life expectancy differs between continents and how it changes over the selected time period. The year slicer makes it possible to examine specific periods and compare changes over time.

- **GDP per Capita Analysis**

The project also analyzes GDP per capita by country and continent. The map and treemap provide a geographical and hierarchical view of economic differences, while the detailed trend page allows users to examine changes in GDP per capita across continents over time.

This helps highlight differences in economic conditions across regions.

3. Interactive Dashboard Features

The Power BI report includes interactive controls that improve the analysis:

- **Year slicer** – allows users to analyze specific years or time periods.
- **Continent slicer** – enables continent-specific analysis.
- **Map visualization** – provides geographical analysis at country level.
- **Donut chart** – shows population distribution across continents.
- **Treemap** – compares countries and continents based on population.
- **Bar chart** – compares average life expectancy across continents.
- **KPI card** – provides a high-level population indicator.
- **Detailed table** – allows country-level comparison of population and life expectancy.
- **Line charts** – show trends in population, life expectancy, and GDP per capita over time.

4. Major Insights

1. Population varies substantially across countries and continents, making geographical analysis important for understanding global population distribution.
2. Life expectancy differs across continents, and the trend analysis allows these differences to be studied over time.
3. GDP per capita shows significant variation across countries, indicating differences in economic conditions and development levels.
4. The combination of GDP per capita and life expectancy allows users to examine the relationship between economic conditions and population well-being.
5. Population trends provide insight into how different regions have changed over the selected years.
6. The year and continent filters make it possible to perform focused comparisons rather than relying only on overall averages.
7. The country-level map and treemap provide a more detailed perspective than continent-level charts, allowing individual countries to be explored.

5. Recommendations

1. Analyze Economic and Social Indicators Together

Use GDP per capita and life expectancy together rather than analyzing them independently. This can provide a broader understanding of differences in economic development and living conditions.

2. Focus on Year-wise Trends

Use the year slicer to compare different periods and identify long-term changes in population, life expectancy, and GDP per capita.

3. Perform Country-Level Comparisons

Countries with significantly different GDP per capita or life expectancy can be compared further to understand regional patterns and differences.

4. Use Interactive Filtering for Deeper Analysis

The continent and year slicers should be used to conduct focused analysis of individual regions and time periods.

5. Enhance the Dashboard with Additional Metrics

If the dataset is extended, additional indicators such as population growth rate, GDP growth, income distribution, healthcare access, and education indicators could provide deeper insights.

Conclusion

The Gapminder Analysis Power BI project transforms global development data into an interactive analytical dashboard covering population, life expectancy, and GDP per capita. The combination of geographical, categorical, tabular, KPI, and time-series visualizations makes it possible to examine global patterns at both country and continent levels.

Overall, the project demonstrates how Power BI can be used to transform multidimensional data into meaningful visual insights and supports comparative analysis of population trends, economic conditions, and life expectancy across different regions and time periods.